

Metric Wedge Obstructions in a Lipschitz-Free Uniform Property (S) Question

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns the Lipschitz-free item in the open-problem section of:

William B. Johnson and Tomasz Kania, *Uniform Property (S)*, arXiv:2602.09106.

The source location is page 19 of the paper.

5. OPEN PROBLEMS

Let us list open problems concerning uniform property (S).

- (1) Does c_0 have equivalent renormings with uniform property (S)?
- (2) Does every space have an equivalent renorming with uniform property (S)?
- (3) Is $U(d, a)$ always smaller than da (up to a constant)?
- (4) Characterise those pointed metric spaces whose Lipschitz-free Banach spaces have uniform property (S). For example, by Rademacher's theorem, $F([0, 1])$ is isometric to $L_1[0, 1]$ so it does have uniform property (S), unlike $F(\mathbb{N})$ that is isometric to ℓ_1 .
- (5) If X has uniform property (S), must X^{**} have it too? The converse follows from the principle of local reflexivity.
- (6) If X and Y are normed spaces and $p \in (1, \infty)$, is there a constant $C_p > 0$ so that the

The paper defines property (S) as follows: a normed space X has property (S) if for every distinct $x, y \in S_X$ and every $a > 0$ there is $z \in X$ with $\|z\| < a$ such that

$$\|x + z\| \neq \|y + z\|.$$

Uniform property (S) is the corresponding quantitative strengthening using the modulus

$$U_X(d; a) = \inf_{\substack{x, y \in S_X \\ \|x - y\| \geq d}} \sup_{\|z\| \leq a} |\|x + z\| - \|y + z\||.$$

Uniform property (S) implies property (S).

Metric Wedges

Let $(A, 0)$ and $(B, 0)$ be pointed metric spaces. Their metric wedge $M = A \vee B$ is the pointed metric space obtained by identifying the two base points and declaring that

$$d(a, b) = d_A(a, 0) + d_B(0, b) \quad (a \in A, b \in B).$$

The original metrics are kept on each piece.

Lemma 1. *For a metric wedge $M = A \vee B$ there is a canonical isometric identification*

$$\mathcal{F}(M) \cong \mathcal{F}(A) \oplus_1 \mathcal{F}(B).$$

Proof. Restriction to the two pieces gives an isometric identification

$$\text{Lip}_0(M) = \text{Lip}_0(A) \oplus_\infty \text{Lip}_0(B).$$

Indeed, if $f_A \in \text{Lip}_0(A)$ and $f_B \in \text{Lip}_0(B)$, the function obtained by gluing f_A and f_B at the base point has Lipschitz constant

$$\max\{\text{Lip}(f_A), \text{Lip}(f_B)\}.$$

The only cross-piece estimate needed is

$$|f_A(a) - f_B(b)| \leq \text{Lip}(f_A)d_A(a, 0) + \text{Lip}(f_B)d_B(b, 0) \leq \max\{\text{Lip}(f_A), \text{Lip}(f_B)\}d(a, b).$$

Taking the canonical predual generated by point evaluations gives $\mathcal{F}(M) \cong \mathcal{F}(A) \oplus_1 \mathcal{F}(B)$. $\square$

Candidate Partial Result

Theorem 1. *Let $M = A \vee B$ be a metric wedge of pointed metric spaces. If A is finite and $|A| \geq 3$, then $\mathcal{F}(M)$ fails property (S), and therefore fails uniform property (S).*

Proof. By the lemma,

$$\mathcal{F}(M) = \mathcal{F}(A) \oplus_1 \mathcal{F}(B).$$

Write $Y = \mathcal{F}(A)$ and $Z = \mathcal{F}(B)$.

Since A is finite, the unit ball of Y is the centrally symmetric polytope

$$B_Y = \text{conv} \left\{ \frac{\delta_u - \delta_v}{d(u, v)} : u, v \in A, u \neq v \right\}.$$

Also $\dim Y = |A| - 1 \geq 2$. Hence B_Y has a positive-dimensional facet. Choose two distinct points x, y in the relative interior of such a facet, and let φ be a norm-one functional exposing that facet.

Because B_Y is a polytope, there are finitely many norm-one functionals $\varphi_1, \dots, \varphi_N$ such that

$$\|u\|_Y = \max_{1 \leq j \leq N} \varphi_j(u) \quad (u \in Y),$$

with one of the φ_j equal to φ . Since x and y lie in the relative interior of the facet exposed by φ , all other facet functionals are strictly smaller than φ at both x and y . By finiteness and continuity there is $\eta > 0$ such that whenever $\|u\|_Y < \eta$,

$$\|x + u\|_Y = \varphi(x + u) \quad \text{and} \quad \|y + u\|_Y = \varphi(y + u).$$

Consequently,

$$\|x + u\|_Y - \|y + u\|_Y = \varphi(x + u) - \varphi(y + u) = \varphi(x) - \varphi(y) = 0$$

for every $\|u\|_Y < \eta$.

Now view x and y as elements of $Y \oplus_1 Z$. They remain distinct unit vectors. If $w = (u, v) \in Y \oplus_1 Z$ and $\|w\| < \eta$, then $\|u\|_Y < \eta$, and therefore

$$\|x + w\| = \|x + u\|_Y + \|v\|_Z = \|y + u\|_Y + \|v\|_Z = \|y + w\|.$$

Thus the pair $x, y \in S_{\mathcal{F}(M)}$ cannot be separated by any perturbation of norm $< \eta$. This violates property (S), and hence also uniform property (S). $\square$

Verification Notes

No computational verifier was used for this packet. The proof is a structural Banach-space argument. The main review points are:

- the standard metric-wedge identity $\mathcal{F}(A \vee B) = \mathcal{F}(A) \oplus_1 \mathcal{F}(B)$;
- the polyhedral finite-free-space obstruction from the finite metric subcase;
- the preservation of the obstruction in an ℓ_1 -sum, because the second coordinate contributes the same term to both perturbed norms.

Limitations and Novelty Check

This packet is only a partial obstruction for the Lipschitz-free open problem. It does not characterize all pointed metric spaces M for which $\mathcal{F}(M)$ has uniform property (S) .

The result strengthens the earlier finite-metric packet only when the second wedge component is nontrivial, especially when it is infinite. The novelty risk is moderate: both ingredients, the metric-wedge decomposition of Lipschitz-free spaces and the finite polyhedral obstruction, are standard. The packet records the combined obstruction as a convenient explicitly checked subcase of the source paper's characterization problem.

The same-source check found no nearby answer to this wedge obstruction in the source paper. The local run registry contains only the finite-metric free-space subcase, not this wedge extension.